

Chengyan (Kelly) Liu

(514) 601-9723 • chengyan.liu@mail.mcgill.ca • [linkedin.com/in/kelly-chengyan-liu/](https://www.linkedin.com/in/kelly-chengyan-liu/) • github.com/CLIUKELLY • [Project Portfolio](#)

EDUCATION

McGill University, Desautels Faculty of Management Montreal, Canada
Master of Management in Analytics Jul 2023 – Dec 2024E

- Holding the entrance scholarship and the Director's Award of \$9k; GPA 3.58/4.0

Concordia University, John Molson School of Business Montreal, Canada
Bachelor of Commerce, Major Business Technology Management, Minor Computer Science with Coop Sep 2016 - May 2021

- Graduated with distinction (GPA 3.51/4.3), Dean's List 2020-2021

SKILLS

Office: MS Office, Visio, Project, SharePoint, Jira, Confluence, Notion

Big Data: Python, SQL, R, Machine Learning (Scikit-learn, SciPy, PyTorch, TensorFlow), NLP (nltk, SpaCy, Vader), GenAI (LLM, Diffusion, GPT4-v), Cloud Services (Azure, AWS), PowerBI, Access, Tableau, Data Federation, Mining, and Governance (Lakehouse, Databricks, Arcteryx, Informatica, IBM COGNOS, IBM Netezza)

Development: Java, Web Frontend, UI/UX Design, Project Management, Solution Architecture Design

PROFESSIONAL EXPERIENCE

Schneider Electric Beijing, China
AI Algorithm Engineer Intern May 2024 — Dec 2024E

- Driving AI Solution Go-to-market: material preparation for sales reps, AI conference preparation and booth keeping
- Assisting with industrial forecasting and optimization solution development for energy reduction and boost efficiency

Foxconn Technology Group (Manufacturer for Apple Products) Chengdu, China
Design for Manufacturing Engineer Jul 2021 — Jul 2022

- Led data driven failure analysis for quality enhancement, yield improvement, and cost-saving for MacBook New Product Introduction Phase, improved process efficiency by more than 50%
- Conducted production simulation to ensure the capacity for the transition of from existing product obsolescence to new product introduction, saved millions USD investment on a new production line by warehouse stacking
- Managed new product line setup project by gathering client needs, driving cross-functional teams to assess plans and find the optimal solution, delegating tasks, and ensuring deliverables are on time and within budget

Novartis Pharmaceuticals Canada Montreal, Canada
Commercial Business Analyst Intern Sep 2018 – Jul 2019 & Jan 2020 – Jun 2020

- Assisted 3 new drug launches, supported brand teams with data insights for strategic planning and problem solving by building forecasting models
- Build dashboards to visualize data from multi-data sources to support brand managers understand market performance, resulting in market share gains and revenue boosts.
- Streamlined report preparation efficiency by creating deliverable preparation templates, semi-automated and optimized scheduled report preparation process

ACADEMIC PROJECTS

CN Rail Data Solution Consulting Project Montreal, Canada
Solution Architect & Strategist & PM Sep 2023 – Apr 2024E

Master of Management in Analytics (MMA) Capstone project in partnership with CN Rail

- Conducting fuel efficiency analysis using data mining techniques, providing energy-saving solutions, and aiming to achieve cost optimization
- Providing data strategic consulting for the company's data division, empowering the company data landscape through the use of data federation, data mesh, and other data strategies
- Utilizing large multimodal models (LMM), AI services, and other cloud services to design, develop, and implement a multimodal document confidentiality classification tool, enhancing classification efficiency through automation

Customer Personality Analysis (Python: Clustering, Causal Inference)*Data Scientist & PM*Montreal, Canada
Jan 2024 – Feb 2024

- Applied clustering and causal inference on customer data for segmentation and determining variable causality, enhancing marketing strategies

"Party Animal" Steam Game Feedback Analysis (Python: NLP, Sentiment Analysis)*NLP Data Scientist*Montreal, Canada
Jan 2024 – Feb 2024

- Analyzed game reviews using NLP and sentiment analysis to support investment strategy optimization

Herbicide Usage Optimization Project (Python: Gurobi, Optimization)*Modeler*Montreal, Canada
Sep 2023 – Dec 2023

- Achieved a reduction in cancer risk-associated herbicide usage by applying combination of mathematical modelling and optimization techniques, successfully upholding crop production and cost targets within treatment method constraints

Motor Vehicle Insurance Risk Analysis (R: PCA, Classification)*Data Scientist*Montreal, Canada
Dec 2023 – Dec 2023

- Conducted insurance risk analysis using PCA and Random Forest models with R and vehicle data, providing driving and purchasing recommendations to stakeholders

Starbucks Promotional Strategy Analysis (Python: Clustering, Classification)*Data Scientist & PM*Montreal, Canada
Sep 2023 – Nov 2023

- Utilized data mining and visualization techniques using Python, analyzed Starbucks consumer behavior and made prediction on offer completion, optimizing promotional strategies to enhanced customer engagement and contributed to revenue growth

Covid-19 Database and Alerting System (SQL, PHP, JavaScript, Web Programming)*Data Engineer & PM*Montreal, Canada
Jan 2021 – May 2021

- Designed and developed a database and alerting system with SQL that records PCR test positive patients and close contacts data by tracking and alerting on positive cases for public health safety management
- Functions Implemented: database maintenance, patients and public health facilities information search, PCR test positive contact alert

Information System Consultation Projects*IT Solution Consultant*Montreal, Canada
Sep 2019 – Dec 2020

Led a series of 3 projects to analyze Novartis Marketing Intelligence Team's information system, identified pain points, and proposed IT strategy, procurement, and management plan

- Project No.1 - Information System Analysis and System Solution Proposal: Assessed client's information system, identified deficiencies, and proposed feasible system improvement plans for work efficiency optimization, information security reinforcement, and information system integration
- Project No.2 - Information System Design and Deployment: Designed and developed a collaborative work tool demo tailored to the client's needs (UI&UX), provided the deployment plan
- Project No.3 - IT strategy, Management, and Sourcing options Analysis: Assessed IT solution based on criteria tailored to needs, assessed and proposed available BI tools with implementation and change management plans

LEADERSHIP AND EXTRACURRICULAR**Canada China Art-Tech (CCAT)***Grant Coordinator*Montreal, Canada
Sep 2017 – Oct 2018

- Prepared 8 grant applications to secure funding for program, receiving \$4K+ in funds and sponsorships
- Assisted with government registrations, financial statement preparation to meet legal reporting requirements of a non-profit organization

OTHER SKILLS AND INTERESTS**Languages:** English (fluent), Mandarin (fluent), French (intermediate)**Interests:** Mix Martial Arts (Boxing, Muay Thai), Fitness